

Work Instruction — Part 34521-A OP30 — Finish Turn + Bore on NLX-08

Rev D · effective 2025-08-05 (aligned with control plan Rev C). Operation OP30 (finish turn OD + bore Ø 38.000 +0.025 / -0.000 mm) on AISI 4140. Backup machine NLX-09 .

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owner: R. Bertolino, Process Engineering - Cleveland
part_numbers: [34521-A]
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machine_ids: [NLX-08, NLX-09]
controller_models: [Fanuc 0i-TF]
fault_codes: []
materials: [MAT-4140]
standards: []
supplier_ids: []
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1. Setup

- 1. Verify machine is warm (warm-up cycle complete — see PDM-MAN-NLX-004 §4).
- 2. Load OP20 part. Chuck pressure 35 bar. Confirm runout < 0.02 mm TIR at OD.
- 3. Load boring bar set BB-34521A-01 (carbide TNMG 16-04 KC5010). Verify in tool table.
- 4. Run program 034521A_OP30_REVD .

2. Cutting parameters

Sub-op	Tool	Spindle (rpm)	Feed (mm/rev)	Depth of cut (mm)	Coolant
2-1: Finish turn OD	T01 — CNMG 12-04 KC5010	1,800	0.20	0.5	15 bar through-tool
2-2: Drill (if not pre-drilled)	T02 — Ø 30 mm insert drill	900	0.08	—	15 bar

2-3: Rough bore	T03 — boring bar BB-34521A-01 rough insert	1,200	0.15	1.5	15 bar (must remain ON throughout pass)
2-4: Finish bore (Ø 38.000 +0.025/-0)	T04 — boring bar BB-34521A-01 finish insert	1,400	0.08	0.2	15 bar (mandatory — see callout)

Coolant must remain ON for the entire finish-bore pass. Per PFMEA QMS-FMEA-34521A-RevB failure mode FM-007, thermal drift from coolant interruption is a top contributor to bore-diameter out-of-tolerance rejections.

3. In-process checks

1. **Every 10 pieces: bore-gauge check (Mitutoyo 511 series).** Required by control plan QMS-CP-34521A-RevC . If bore Ø trends > +0.015 mm from nominal, change finish insert immediately and re-verify on next piece.
2. Every setup: profilometer check, $R_a \leq 0.8 \mu\text{m}$.
3. Insert change at 200 pieces (mandatory) or sooner per gauge trend.

4. Rejection handling

If bore diameter measured at this op falls outside Ø 38.000 +0.025 / –0.000 mm, **stop and follow control plan reaction plan** (QMS-CP-34521A-RevC §3). Failure mode is FM-007 in QMS-FMEA-34521A-RevB . Do not adjust tool offset and continue running.

5. Safety and PPE

- Cutting fluid is EcoCut 7700 — see SDS ERP-SDS-ECOCUT7700 .
- Chips are sharp ribbons — use chip hook, never bare hands.
- Door interlock active during cutting. Chuck rotation key removed from controller when reaching into chuck area.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	R. Bertolino, Process Engineering — Cleveland	(on file)	2025-08-05
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